INCH POUND

MIL-DTL-60517C
25 March 2011
SUPERSEDING
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14 September 2004

DETAIL SPECIFICATION

# INITIATOR, CARTRIDGE ACTUATED, M27 METAL PARTS ASSEMBLY

*This specification is approved for use by all Departments and Agencies of the Department of Defense.*

## 1. SCOPE

**1.1 Scope.** This specification covers the requirements for the M27 Cartridge Actuated Initiator Metal Parts Assembly (see 6.1).

## 2. APPLICABLE DOCUMENTS

**2.1 General.** The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

**2.2 Government documents.**

**2.2.1 Specifications, standards, and handbooks.** The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

| | |
|---|---|
| MIL-A-2550 | Ammunition, General Specification for |

(Copies of these documents are available online at https://assist.daps.dla.mil/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

Comments, suggestions, or questions on this document should be addressed to DEPARTMENT OF THE NAVY, Indian Head Division, NSWC, Code E12AP, Document Control, 4123 Artisans Court, Suite 103, Indian Head, Maryland 20640-5085, OFFICIAL BUSINESS, or emailed to amanda.penn@navy.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.daps.dla.mil/.

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**2.2.2 Other Government documents, drawings, and publications.** The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DRAWINGS

NAVAL ORDNANCE STATION, INDIAN HEAD (GAGE CODE 19200 and 19203)

| | |
|---|---|
| 10520551 | Initiator, Cartridge Actuated, M27, Metal Parts Assembly |
| 8830984 | Box, Fiberboard (inner) packing, Ammunition for Initiator, Cartridge Actuated M27, M86, M87, M99, or XM107 |
| 8830985 | Box, Packing, Ammunition, for Initiator, Cartridge Actuated M27, M86, M87, M98, M99, or XM017 |
| 9241155 | Box, Fiberboard (Outer) Alternative Packing, Ammunition, for Initiator, Cartridge Actuated M27, M86, M87, M98, M99, or XM107 |

(Application for copies should be addressed to Commanding Officer, Naval Surface Warfare Center, Indian Head, Technical Information Branch (Code 4230), Indian Head, MD 20640.)

**2.3 Non-Government publications.** The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY FOR QUALITY

| | |
|---|---|
| ASQ A8402 | Quality Management and Quality Assurance – Vocabulary |
| ASQ Z1.4 | Sampling Procedures and Tables for Inspection by Attributes |

(Application for copies should be addressed to the American Society for Quality Control, PO Box 3005, 611 E. Wisconsin Ave., Milwaukee, WI 53201-4606.)

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION

| | |
|---|---|
| ISO 3601-3 | Fluid power systems — O-ring — Part 3: Quality acceptance criteria |

(Application for copies should be addressed to the International Organization for Standardization, 1, ch. de la Voie-Creuse, Case postale 56 CH-1211 Geneva 20, Switzerland)

**2.4 Order of precedence.** Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

## 3. REQUIREMENTS

**3.1 First Article.** When specified in the contract or purchase order (see 6.2), a sample shall be subjected to first article inspection (see 4.4 and 6.3).

**3.2 Materials.** All materials used in the manufacture of the initiators shall conform strictly to the specifications referred to on the respective drawings unless specific approval in writing covering a departure there from has been obtained from the cognizant Navy design activity prior to manufacture.



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When alternate materials or methods of manufacture are specified on the drawings, the bidder's selections shall be clearly stated in the proposal.

**3.2.1 Raw materials.** The contractor shall be responsible for selecting from the classification list contained in the raw material specification the surface and physical condition of the raw material intended for use in manufacture.

**3.3 Hardness and other physical properties.** Physical properties required by the drawing shall be uniform throughout the part within specified tolerances. Requirements for all alloy steel parts shall be achieved by quenching and tempering.

**3.4 Hydrostatic pressure.** The initiator chamber and cap shall withstand a hydrostatic pressure of 14,000 to 15,000 pounds per square inch (psi) for 15 seconds, minimum, without deformation or mechanical failure (see 6.4.1).

**3.5 Firing pin release.** The firing pin of the metal parts assembly initiator, less safety pin and cartridge, shall release from the cocked position when the application of an axial load of 15 to 30 pounds has withdrawn the initiator pin 0.370 inch, minimum, and 0.425 inch, maximum (see 4.3.3).

**3.6 Initiator pin security.** Application of an axial load of 200 to 220 pounds to the initiator pin of the metal parts assembly shall not withdraw the initiator pin past the safety pin. Load application shall cause no permanent deformation of the safety pin.

**3.7 Protective finish.** Protective finish shall be in accordance with the applicable specifications and shall apply to all surfaces of the component regardless of contour configuration.

**3.8 Residual magnetism.** The metal parts assembly shall not deflect the indicator of the compass more than five degrees in either direction using procedures in 4.6.6.

**3.9 Metal parts assembly.** Parts shall be combined to produce an assembly, which conforms to the requirements of the metals parts assembly drawing. All threaded joints shall be assembled hand tight.

**3.10 Workmanship.** The requirements for workmanship are as specified on the applicable drawings, referenced specification, and the following:

**3.10.1 Metal defects.** All components shall be free from cracks, splits, cold shuts, inclusions, porosity, or any similar defect.

**3.10.2 Burr.** No part shall have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

**3.10.3 Foreign matter.** No part or assembly shall contain chips, dirt, grease, rust, corrosion, or other foreign matter.

## 4. VERIFICATION

**4.1 Classification of inspections.** The inspection requirements specified herein are classified as follows:

a. Conformance inspection.



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**4.1.1 Quality assurance terms and definitions.** Reference shall be made to ASQ A8402 to define the quality assurance terms used.
**4.1.2 Inspection.** Inspection shall be in accordance with MIL-A-2550.

**4.2 Classification of Characteristics.** The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4)
b. Quality conformation inspection (see 4.5)

**4.3 Inspection conditioning.** Unless otherwise specified (see 6.2), all inspections shall be performed under the following conditions:

a. Temperature: Room ambient 20 to 25 °C (68 to 77 °F)
b. Altitude: Sea level
c. Humidity: 50 percent relative, maximum
d. Atmospheric pressure: 14.7 psia.

**4.4 First article inspection.** At the beginning of regular production, the supplier shall submit a first article sample to a Government approved facility for evaluation. The sample shall consist of 5 metal parts assemblies, 9 sets of unassembled parts, and 1 set of unassembled parts without plating or final protective finish. The sample shall be manufactured in the same manner, using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, must be the same as will be used in regular production and shall be obtained from the same source of supply. After inspection and provisional acceptance at source, the first article sample shall be inspected for all requirements of the drawings and specifications. The Government reserves the right to require new first article samples until such time as an acceptable sample is submitted (see 6.3).

**4.5 Quality conformance inspection.**

**4.5.1 Lot.**

**4.5.1.1 Submission of product.** The product shall be submitted in accordance with ASQ Z1.4.

**4.5.1.2 Size of lot.** For the purpose of identification and shipment, the shipping lots of metal parts assemblies shall consist of 1200 assemblies, maximum, provided the homogeneity requirements of inspection lots have been retained.

**4.5.2 Examination.** One-hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein. Four power (4X) magnification may be used in performing visual examination for such defects as burr and foreign matter. Inspection for rubber O-rings shall be in accordance with ISO 3601-3. When compliance with the applicable requirements is in doubt as a result of visual examination, the characteristic shall be measured or gaged to determine acceptability. All nonconforming material shall be rejected.



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**4.5.2.1 Chamber, Initiator, Drawing 10520407, covering a detail of Drawing 10520551.**

| Characteristic | Defect | Method of Inspection |
|---|---|---|
| **Critical** | | |
| None Defined | | |
| **Major** | | |
| 101 | Pitch diameter of large external thread, min | Gage |
| 102 | Major diameter of large external thread, min | Gage |
| 103 | Pitch diameter of small external thread, min | Gage |
| 104 | Major diameter of small external thread, min | Gage |
| 105 | Pitch diameter of internal thread, max | Gage |
| 106 | Minor diameter of internal thread, max | Gage |
| 107 | Concentricity of datum location with pitch diameter of small external thread | Gage |
| 108 | Concentricity of pitch diameter of large thread with cavity | Gage |
| 109 | Concentricity of large external thread pilot with large external thread | Gage |
| 110 | Concentricity of counterbore with cavity, including perpendicularity with base of counterbore | Gage |
| 111 | Diameter of counterbore | Gage |
| 112 | Diameter of large thread pilot | Gage |
| 113 | Depth of cavity | Gage |
| 114 | Length of datum location | Gage |
| 115 | Thickness of wall, min | Gage |
| 116 | Metal defective | Visual |
| 117 | Protective finish, inadequate or defective | Visual |
| 118 | Surface finish of cavity and angle of gas port, improper | Visual |
| 119 | Burr | Visual |
| 120 | Foreign matter present | Visual |
| **Minor** | | |
| 201 | Pitch diameter of large external thread, max | Gage |
| 202 | Pitch diameter of small external thread, max | Gage |
| 203 | Pitch diameter of internal thread, min | Gage |
| 204 | Diameter of small external relief | Gage |
| 205 | Diameter of small external thread pilot | Gage |
| 206 | Diameter of hole in gas port | Gage |
| 207 | Diameter of cavity | Gage |
| 208 | Depth of counterbore | Gage |
| 209 | Length of gas port | Gage |
| 210 | Length of small thread relief | Gage |
| 211 | Length from face of gas port to small thread | Gage |
| 212 | Length of external thread pilot | Gage |
| 213 | Thickness of wall, max | Gage |
| 214 | Depth of threaded hole | Measure |
| 215 | Diameter of chamfer at internal thread | Measure |
| 216 | Total length | Measure |
| 217 | Length of large external thread | Measure |
| 218 | Marking missing, incorrect or illegible | Visual |



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| | | |
|---|---|---|
| 219 | Radius or chamfer, missing or incorrect | Visual |
| 220 | Surface finish improper, except as otherwise classified | Visual |

**4.5.2.2 Cap, Initiator, Drawing 10520552, covering a detail of Drawing 10520551.**

**Critical**

None defined

**Major**

| | | |
|---|---|---|
| 101 | Pitch diameter of small internal thread, max | Gage |
| 102 | Minor diameter of small internal thread, max | Gage |
| 103 | Pitch diameter of large external thread, max | Gage |
| 104 | Minor diameter of large external thread, max | Gage |
| 105 | Concentricity of pitch diameter of large internal thread with midsection cavity, including perpendicularity with base of counterbore of large cavity | Gage |
| 106 | Concentricity of pitch diameter of small internal thread with midsection cavity, including perpendicularity with base of counterbore of large cavity | Gage |
| 107 | Concentricity of hole between small cavity and boss cavity with midsection cavity, including perpendicularity with base of counterbore of large cavity | Gage |
| 108 | Concentricity of undercut in large cavity with large internal thread | Gage |
| 109 | Concentricity of boss cavity with hole between small cavity and boss cavity | Gage |
| 110 | Concentricity of flange with hole between small cavity and boss cavity | Gage |
| 111 | Concentricity of body, large end with large internal thread | Gage |
| 112 | Concentricity of small cavity with midsection cavity | Gage |
| 113 | Concentricity of thread with large internal thread | Gage |
| 114 | Symmetry of thickness of mounting lugs with body, large end | Gage |
| 115 | Symmetry of length of mounting lugs with body, large end | Gage |
| 116 | Variation of centerline of mounting holes with small hole between boss cavity and small cavity | Gage |
| 117 | Diameter of large cavity rear of undercut | Gage |
| 118 | Diameter of small cavity | Gage |
| 119 | Diameter of boss cavity, max | Gage |
| 120 | Diameter of hole between small cavity and boss cavity, max | Gage |
| 121 | Diameter of undercut at large internal thread | Gage |
| 122 | Diameter of safety pin hole, max | Gage |



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| | | |
|---|---|---|
| 123 | Diameter of flange | Gage |
| 124 | Length of effective small interior thread, min | Gage |
| 125 | Length of small interior thread, max | Gage |
| 126 | Width of undercut behind large interior thread | Gage |
| 127 | Location of safety pin hole | Gage |
| 128 | Height of boss | Gage |
| 129 | Metal defective | Visual |
| 130 | Protective finish, inadequate or defective | Visual |
| 131 | Burr | Visual |
| 132 | Foreign matter present | Visual |
| **Minor** | | |
| 201 | Pitch diameter of set screw thread | Gage |
| 202 | Minor diameter of set screw thread | Gage |
| 203 | Pitch diameter of small internal thread, min | Gage |
| 204 | Pitch diameter of large internal thread, min | Gage |
| 205 | Diameter of body, small end | Gage |
| 206 | Diameter of body, large end | Gage |
| 207 | Diameter of groove, small end | Gage |
| 208 | Diameter of chamfer, large internal thread | Gage |
| 209 | Diameter of chamfer, small internal thread | Gage |
| 210 | Diameter of midsection cavity | Gage |
| 211 | Diameter of boss cavity, min | Gage |
| 212 | Diameter of hole between boss cavity and small cavity, max | Gage |
| 213 | Diameter of safety pin hole, min | Gage |
| 214 | Diameter of mounting hole | Gage |
| 215 | Depth from base of large cavity rear of undercut to base midsection cavity | Gage |
| 216 | Depth from base of large cavity rear undercut to face of internal boss | Gage |
| 217 | Depth to bottom of large cavity rear of undercut | Gage |
| 218 | Depth of large cavity rear of undercut | Gage |
| 219 | Total length | Gage |
| 220 | Length of body, large end | Gage |
| 221 | Width of groove, small end | Gage |
| 222 | Width across flat | Gage |
| 223 | Thickness of mounting flange | Gage |
| 224 | Location of set screw hole | Gage |
| 225 | Location of groove, small end | Gage |
| 226 | Location of small cavity | Gage |
| 227 | Location of mounting holes | Gage |
| 228 | Location of flats | Gage |
| 229 | Length from rear flange to shoulder | Measure |
| 230 | Width of flange | Measure |
| 231 | Safety pin hole obstructed | Visual |
| 232 | Marking missing, incorrect or illegible | Visual |
| 233 | Radius or chamfer, missing or incorrect | Visual |
| 234 | Surface finish, improper | Visual |



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**4.5.2.3 Pin, Initiator, Drawing 10520553, covering a detail of Drawing 10520551.**

| | | |
|---|---|---|
| **Critical** | | |
| None Defined | | |
| **Major** | | |
| 101 | Concentricity of groove in head with body | Gage |
| 102 | Concentricity of shoulder with body | Gage |
| 103 | Symmetry of pull hole with body | Gage |
| 104 | Diameter of undercut, min, if applicable | Gage |
| 105 | Width of undercut, max, if applicable | Gage |
| 106 | Diameter of body | Gage |
| 107 | Diameter of shoulder | Gage |
| 108 | Diameter of groove in body, min | Gage |
| 109 | Width of head groove | Gage |
| 110 | Location of groove in head | Gage |
| 111 | Location of groove in body | Gage |
| 112 | Location of pull hole | Gage |
| 113 | Profile of end | Visual |
| 114 | Profile of head | Visual |
| 115 | Metal defective | Visual |
| 116 | Protective finish, inadequate or defective | Visual |
| 117 | Surface finish of groove in head and face of shoulder, improper | Visual |
| 118 | Burr | Visual |
| 119 | Foreign matter present | Visual |
| **Minor** | | |
| 201 | Diameter of groove in head | Gage |
| 202 | Diameter of groove in body | Gage |
| 203 | Diameter of pull hole | Gage |
| 204 | Length to groove in body | Gage |
| 205 | Width of flat at body groove end | Gage |
| 206 | Thickness of pull hole end | Gage |
| 207 | Total length | Measure |
| 208 | Marking missing, incorrect or illegible | Visual |
| 209 | Radius or chamfer, missing or incorrect | Visual |
| 210 | Surface finish, improper, except as otherwise classified | Visual |

**4.5.2.4 Seal Drawing 10520415, covering a detail of Drawing 10520551.**

| | | |
|---|---|---|
| **Critical** | | |
| None defined | | |
| **Major** | | |
| 101 | Diameter, outside, min | Gage |
| 102 | Width, min | Gage |
| 103 | Thickness, min | Gage |
| 104 | Cut, cracked or split | Visual |
| 105 | Flash | Visual |
| 106 | Foreign matter present | Visual |
| **Minor** | | |
| 201 | Diameter, outside, max | Gage |



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| | | |
|---|---|---|
| 202 | Width, max | Gage |
| 203 | Thickness, max | Gage |

**4.5.2.5 Housing, Firing, Pin, Drawing 10520410, covering a detail of Drawing 10520551.**

| | | |
|---|---|---|
| **Critical** | | |
| 1 | Diameter of striker hole, min | Gage |
| **Major** | | |
| 101 | Pitch diameter of thread, min | Gage |
| 102 | Major diameter of thread, min | Gage |
| 103 | Concentricity of striker hole with cavity | Gage |
| 104 | Concentricity of cavity with body | Gage |
| 105 | Concentricity of pitch diameter of thread with body | Gage |
| 106 | Perpendicularity of face with body | Gage |
| 107 | Diameter of body, min | Gage |
| 108 | Diameter of cavity | Gage |
| 109 | Diameter of flange, min | Gage |
| 110 | Diameter of boss | Gage |
| 111 | Diameter of undercut, min | Gage |
| 112 | Total length, max | Gage |
| 113 | Width of undercut | Gage |
| 114 | Thickness of striker hole | Gage |
| 115 | Location of thread | Gage |
| 116 | Chamfer at striker hole | Gage |
| 117 | Height of boss | Visual |
| 118 | Metal defective | Visual |
| 119 | Protective finish, inadequate or defective | Visual |
| 120 | Surface finish of cavity, improper | Visual |
| 121 | Burr | Visual |
| 122 | Foreign matter present | Visual |
| **Minor** | | |
| 201 | Pitch diameter of thread, max | Gage |
| 202 | Diameter of body, max | Gage |
| 203 | Diameter of striker hole, max | Gage |
| 204 | Diameter of flange, max | Gage |
| 205 | Diameter of wrench holes | Gage |
| 206 | Depth of wrench holes | Gage |
| 207 | Total length, min | Gage |
| 208 | Length of body to flange | Gage |
| 209 | Location of wrench holes | Gage |
| 210 | Marking missing, incorrect or illegible | Visual |
| 211 | Radius or chamfer, missing or incorrect, except as otherwise classified | Visual |
| 212 | Surface finish, improper, excepts as otherwise classified | Visual |



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**4.5.2.6 Pin, Firing, Drawing 10520411, covering a detail of Drawing 10520551.**

| | | |
|---|---|---|
| **Critical** | | |
| None Defined | | |
| **Major** | | |
| 101 | Concentricity of large cavity with body | Gage |
| 102 | Concentricity of striker with body | Gage |
| 103 | Concentricity of cavity with body | Gage |
| 104 | Symmetry of pin relief holes with small cavity | Gage |
| 105 | Symmetry of ball hole counterbore with ball holes | Gage |
| 106 | Diameter of striker | Gage |
| 107 | Diameter of body | Gage |
| 108 | Diameter of ball hole, max | Gage |
| 109 | Diameter of ball hole counterbore | Gage |
| 110 | Diameter of pin relief holes, min | Gage |
| 111 | Depth of ball counterbore, min | Gage |
| 112 | Length of striker, min | Gage |
| 113 | Location of ball hole counterbore | Visual |
| 114 | Metal defective | Visual |
| 115 | Protective finish, inadequate or defective | Visual |
| 116 | Surface finish of body, improper | Visual |
| 117 | Burr | Visual |
| 118 | Foreign matter present | Visual |
| **Minor** | | |
| 201 | Diameter of large cavity | Gage |
| 202 | Diameter of small cavity | Gage |
| 203 | Diameter of ball hole, min | Gage |
| 204 | Diameter of pin relief holes, max | Gage |
| 205 | Depth of large cavity | Gage |
| 206 | Depth to bottom of small cavity | Gage |
| 207 | Depth of ball hole counterbore, max | Gage |
| 208 | Length of striker, max | Gage |
| 209 | Total length | Gage |
| 210 | Location of pin relief holes | Gage |
| 211 | Marking missing, incorrect or illegible | Visual |
| 212 | Radius or chamfer, missing or incorrect | Visual |
| 213 | Surface finish, improper, except as otherwise classified | Visual |

**4.5.2.7 Spring, Initiator, Drawing 10520412, covering a detail of Drawing 10520551.**

| | | |
|---|---|---|
| **Critical** | | |
| None defined | | |
| **Major** | | |
| 101 | Load at specified height | Gage |
| 102 | Closed length, max | Gage |
| 103 | Ends incorrect | Visual |
| 104 | Protective finish, inadequate or defective | Visual |
| 105 | Burr | Visual |
| 106 | Foreign matter present | Visual |



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**Minor**

| | | |
|---|---|---|
| 201 | Diameter outside | Gage |
| 202 | Free length | Gage |

**4.5.2.8 Pin, Safety, Drawing 10522817, covering a detail of Drawing 10520551.**

**Critical**

None defined

**Major**

| | | |
|---|---|---|
| 101 | Diameter of wire | Gage |
| 102 | Profile, incorrect | Visual |
| 103 | Metal defective | Visual |
| 104 | Protective finish, inadequate or defective | Visual |
| 105 | Burr | Visual |
| 106 | Foreign matter present | Visual |

**Minor**

| | | |
|---|---|---|
| 201 | Diameter of head, outside | Measure |
| 202 | Length between high and low ends | Measure |
| 203 | Total length | Measure |
| 204 | Location of center loop | Measure |
| 205 | Radius of center loop, incorrect | Measure |

**4.5.2.9 Filter, Drawing 10520414, covering a detail of Drawing 10520551.**

**Critical**

None Defined

**Major**

| | | |
|---|---|---|
| 101 | Pitch diameter of thread, min | Gage |
| 102 | Major diameter of thread, min | Gage |
| 103 | Concentricity of thread with center hole and outside diameter | Gage |
| 104 | Symmetry of hole with outside diameter | Gage |
| 105 | Diameter of center hole | Gage |
| 106 | Diameter of holes | Gage |
| 107 | Depth of center hole | Gage |
| 108 | Depth of slot, max | Gage |
| 109 | Location of holes | Gage |
| 110 | Angular location of holes | Gage |
| 111 | Nylon patch not properly located or fused | Manual/Visual |
| 112 | Number of holes, incorrect | Visual |
| 113 | Metal defective | Visual |
| 114 | Protective finish, inadequate or defective | Visual |
| 115 | Burr | Visual |
| 116 | Foreign matter present | Visual |

**Minor**

| | | |
|---|---|---|
| 201 | Pitch diameter of thread, max | Gage |
| 202 | Diameter of undercut | Gage |
| 203 | Diameter of body | Gage |
| 204 | Depth of slot, min | Gage |
| 205 | Total length | Gage |
| 206 | Length of body | Gage |



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| | | |
|---|---|---|
| 207 | Width of undercut, max | Gage |
| 208 | Width of slot | |
| 209 | Location of slot | Gage |
| 210 | Marking missing, incorrect or illegible | Visual |
| 211 | Chamfer, missing or incorrect | Visual |
| 212 | Surface finish, improper | Visual |

**4.5.2.10 Assembly, Drawing 10520551, less all drawings except 10520410, 10520411, 10520552, and MS19060-3.**

| | | |
|---|---|---|
| **Critical** | | |
| 1 | Firing pin protrusion | Gage |
| 2 | Steel ball missing | Gage |
| **Major** | | |
| 101 | Burr | Visual |
| 102 | Foreign matter present | Visual |
| **Minor** | | |
| None defined | | |

**4.5.5.11 Assembly, Drawing 10520551.**

| | | |
|---|---|---|
| **Critical** | | |
| None defined | | |
| **Major** | | |
| 101 | Firing pin cocked | Manual |
| 102 | Initiator pin not free to rotate | Manual |
| 103 | Any part missing, except as otherwise classified | Visual |
| 104 | Improper assembly | Visual |
| 105 | Safety pin improperly inserted | Visual |
| 106 | O-ring not lubricated | Visual |
| 107 | Marking missing, incorrect or illegible | Visual |
| **Minor** | | |
| 201 | Protective cap missing | Visual |
| 202 | Protective finish inadequate or defective | Visual |

**4.5.2.12 Box, Fiberboard (inner Ammunition Packing, Drawing 8830984.**

| | | |
|---|---|---|
| **Critical** | | |
| None defined | | |
| **Major** | | |
| None defined | | |
| **Minor** | | |
| 201 | Contents loose | Manual |
| 202 | Box damaged | Visual |
| 203 | Sealing strip missing, incorrectly applied | Visual |
| 204 | Marking missing, incorrect or illegible | Visual |
| 205 | Cut damaged or improperly sealed bag | Visual |



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**4.5.2.13 Box, Ammunition, Packing, Drawing 8830985.**

| | | |
|---|---|---|
| **Critical** | | |
| None defined | | |
| **Major** | | |
| None defined | | |
| **Minor** | | |
| 201 | Contents loose | Manual |
| 202 | Strapping missing, loose, broken, or improperly assembled | Manual/Visual |
| 203 | Contents of box exposed | Visual |
| 204 | Box damaged | Visual |
| 205 | Marking missing, incorrect or illegible | Visual |
| 206 | Number of units incorrect | Visual |

**4.5.2.14 Box, Fiberboard (Outer) Alternative, Ammunition Packing, Drawing 9241155.**

| | | |
|---|---|---|
| **Critical** | | |
| None defined | | |
| **Major** | | |
| None defined | | |
| **Minor** | | |
| 201 | Contents loose | Manual |
| 202 | Box broken or damaged | Visual |
| 203 | Number of units contained incorrect | Visual |
| 204 | Marking missing, incorrect or illegible | Visual |
| 205 | Tape incomplete | Visual |

**4.5.3 Inspection testing.** The following tests shall be performed in accordance with the provision of 4.6.

**4.5.3.1 Hardness.** Fifty units from each heat-treated lot of components shall be tested for compliance with the requirements on the applicable drawing. Failure of any unit to comply with the drawing requirements shall be cause for rejection of the lot of components.

**4.5.3.2 Hydrostatic pressure.** Each chamber and cap shall be subjected to the hydrostatic pressure test. Failure of any cap or chamber subassembly to comply with the requirements of 3.4 shall be cause for rejection of the subassembly.

**4.5.3.3 Firing pin release.** The initiator pin of each partial assembly, consisting of all parts except the safety pin, cartridge and chamber, shall be subjected to the firing-pin release test. Failure of any partial assembly to comply with the requirements of 3.5 shall be cause for rejection of the subassembly.

**4.5.3.4 Initiator pin security.** The initiator pin of each subassembly, consisting of all parts except the chamber, shall be subjected to the initiator pin security test. Failure of any subassembly to comply with the requirements of 3.6 shall be cause for rejection of the subassembly.

**4.5.3.5 Protective finish.** The number of units for protective finish tests shall be in accordance with the applicable specification. Failure of any unit in the inspection sample to comply with the requirements of the applicable specification shall be cause for rejection of the lot.



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**4.5.3.6 Residual magnetism.** Each assembly shall be subjected to the test for excessive residual magnetism. Failure of any unit in the inspection sample to comply with the requirements of 3.8 shall be cause for rejection of assembly.

**4.6 Test methods and procedures.**

**4.6.1 Hardness.** Prior to the application of protective finish, a hardness tester shall be used to determine compliance with the requirements of the applicable drawing.

**4.6.2 Hydrostatic pressure.** Prior to the application of protective finish, the cap and chamber shall be mounted in the device, using a fixture for properly sealing all openings and subjected to the internal hydrostatic pressure to determine compliance with the requirements of 3.4.

**4.6.3 Firing pin release.** The subassembly shall be secured in the fixture. A pull shall be applied to the initiator pin in the longitudinal direction in conjunction with the fixture, which registers the amount of pull.

**4.6.4 Initiator pin security.** The subassembly shall be secured in the fixture with the safety pin in place. A pull shall be applied to the initiator pin in the longitudinal direction in conjunction with a fixture, which registers the amount of pull. The test shall be performed with the safety pin in both positions.

**4.6.5 Protective finish.** Test procedures for protective finish shall be in accordance with the applicable specification referenced on the component drawing.

**4.6.6 Residual magnetism.** The assembly shall be tested for excessive residual magnetism using an approved compass in an area free of local magnetic effects by placing the assembly 5 inches from the end and at the same height as the compass, in a north-south horizontal position (cap end at south) with the compass set in an east-west heading. The assembly shall be moved at a speed no greater the 10 feet per minute past the compass for its entire length, rotated 90 degrees in the horizontal position about its longitudinal axis and moved slowly back past the compass. This procedure shall be repeated for each 90-degree rotation of the assembly for 4 full passes.

**4.7 Inspection of packaging.** The sampling and inspection of the packing and container marking shall be in accordance with the requirements of section 5.

## 5. PACKAGING

**5.1 Packaging.** For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2 and 6.4). When packaging of materiel is to be performed by DOD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Service or Defense Agency, or within the military service's system command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

## 6. NOTES

(This section contains information of a general or explanatory nature that may helpful but is not mandatory.)



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**6.1 Intended use.** The assembly will ultimately be loaded and used in conjunction with other propellant actuated devices in aircrew emergency escape systems. The assembly was designed for use in military aircraft and has no commercial application.

**6.2 Acquisition requirements.** Acquisition documents should specify the following:

a. Title, number, and date of the specification.
b. If required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. If first article is necessary (see 3.1 and 6.3)
d. The responsible for inspection activity (see 4.1)
e. Any special inspection conditions (see 4.3)
f. Level of marking and packing (see 5.1 and 6.4)

**6.3 First article.** When a first article inspection is required, the terms should be a first article sample. The first article should consist of 5 metal parts assemblies, 9 sets of unassembled parts, 1 set of unassembled parts without plating or final protective finish. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first article. Invitations for bids should provided that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

**6.4 Contract packaging, packing and marking.** The following requirements should be specified in the contract.

**6.4.1 Packing and marking level.** Packing and marking should be Level A or B, as specified by the contracting activity.

**6.4.1.1 Level A.** Level A should be in accordance with Drawings 8830984 and 8830985 or SPI F00-138-3770.

**6.4.1.2 Level B.** Level B should be in accordance with Drawings 8830984 and 9241155 or SPI F00-138-3770.

**6.4.1.3 Lot numbers.** Lot numbers should be in accordance with MIL-STD-1168.

**6.5 Definitions.**

**6.5.1 Mechanical failure.** Mechanical failure is any deformation or breakage of a part, the occurrence of which is other than a design function.

**6.6 Subject term (key word) listing.**

Ejection systems, aircrew
Emergency escape systems, aircrew
Escape systems, aircrew
Life-saving systems, aircrew



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**6.7 Changes from previous issue.** Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes. These changes were authorized by Indian Head ECP 11E25CP001.

Custodian:
Navy-OS
Air Force-99

Preparing activity
Navy - OS
(Project 1377-2011-011)

Review activity:
Air Force-70

**NOTE:** The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.daps.dla.mil.



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